

Sr.No	Module	Weightage %	Marks Weightage%
1	Module-II	20.00	26.32
2	Module-III	20.00	26.32
3	Module-I	20.00	25.00
4	Module-IV	20.00	14.47
5	Module-V	20.00	7.89

Weightage representation okay

Weightage representation exceeded more than 5%

Weightage representation exceeded more than 10%

CSE601

[ET]

Enroll. No. -

END SEMESTER EXAMINATION : April-May, 2026

Data Structures and Algorithm Design

Time : 3.00 Hrs

Maximum Marks : 60

Note: Attempt questions from all sections as directed.

Section - A : Attempt any Four questions out of Five . Each question carries 06 marks.

Q1.

(a)

What are Time Complexity and Space Complexity?

(2)

Delete

(b)

Define a Circular Doubly Linked List. Give an example diagram.

(2)

Delete

(c)

Define the Merge Sort Algorithm.

(2)

Delete

Q2.

(a)

Define AVL Tree.

(2)

Delete

(b)

Define the Balancing Factor in the AVL Tree.

(2)

Delete

(c)

Define a binary search tree.

(2)

Delete

Q3.

(a)

Convert Postfix expression of the expression (A+B)*(C-C) and (a+b) * (c+d)/f

(4)

Delete

(b)

State the importance of Dynamic Programming

(2)

Delete

Q4.

(a)

Define an Expression Tree with an example.

(3)

Delete

(b)

Define String Matching Algorithm and Approximation Algorithm.

(3)

Delete

Q5.

When a Problem in NP is NP-Complete? Give an example.

(6)

Delete

Section – B : Attempt any two questions out of three. Each question carries 10 marks.

Q6.

(a)

Develop Java Code for Level Order Traversal in a General Binary Tree.

(5)

Delete

(b)

Discuss the different rotations in the AVL Tree with an example. Write code snippets for each rotation.

(5)

Delete

Q7.

(a)

Develop a complete Java Program for Circular Queue operations.

(5)

Delete

(b)

Discuss the Merge sort algorithm analysis with example data.

(5)

Delete

Q8.

(a)

Develop Java code to find the Nth Fibonacci term using Dynamic Programming.

(5)

Delete

(b)

Define Minimum Spanning Tree (MST), and write the Kruskal Algorithm to find the Minimum Spanning Tree (MST) of a given Graph.

(5)

Delete

Section - C : Compulsory question

Q9.

(a)

Develop Java Code to insert a node at the end of the given Singly Linked List head.

(4)

Delete

(b)

Develop Java code for Push() and Pop() operations.

(4)

Delete

(c)

Discuss Recursion vs Dynamic Programming.

(4)

Delete

(d)

Write the Time Complexity of all the sorting algorithms.

(4)

Delete